



Image Deblurring

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Problem definition

Blurred image



Latent image



Blur kernel

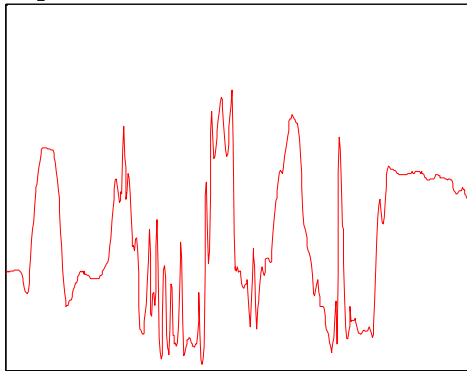


$$\mathbf{B} = \mathbf{I} \otimes \mathbf{K} + \mathbf{n}$$

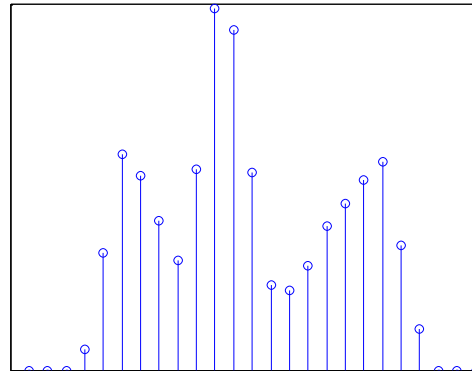
Deblurring: Inverse problem to solve latent $\mathbf{I}$ from given observed noisy $\mathbf{B}$ with (*non-blind*) or without (*blind*) the kernel $\mathbf{K}$

Simple example

Spatial domain

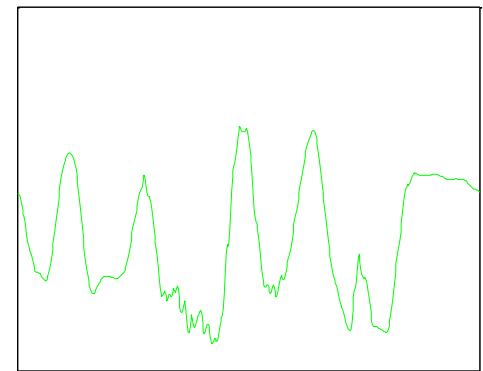


I

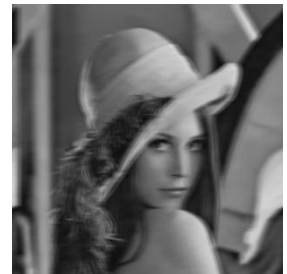


K

$=$

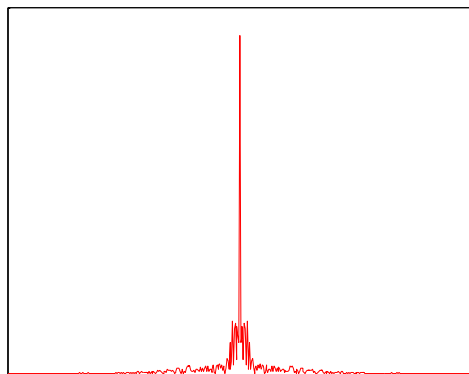


B



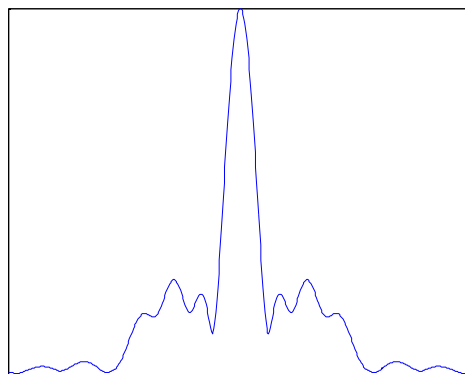
Simple example

DFT domain (magnitude)



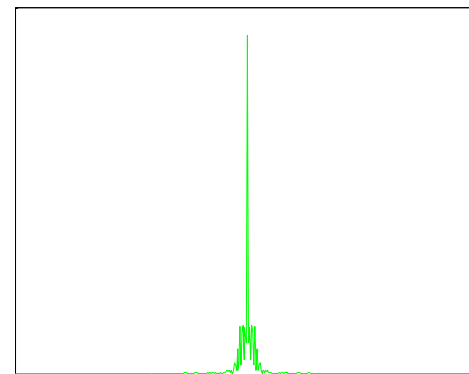
I

$\times$

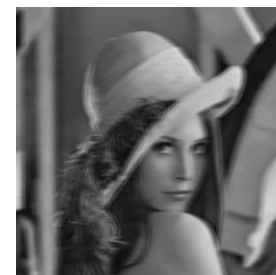
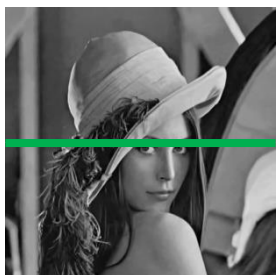


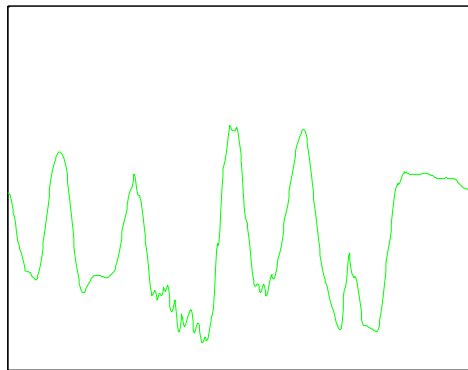
K

$=$

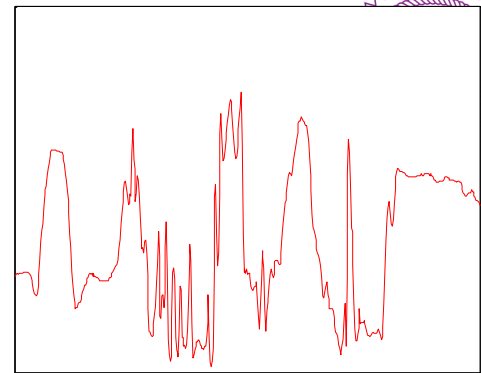


B

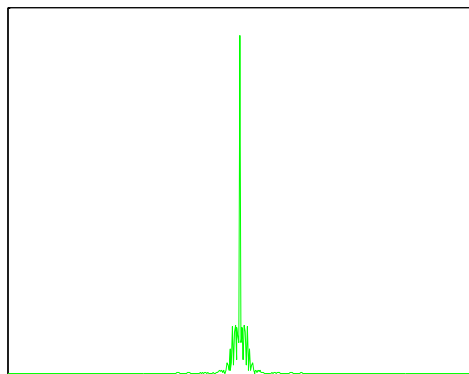




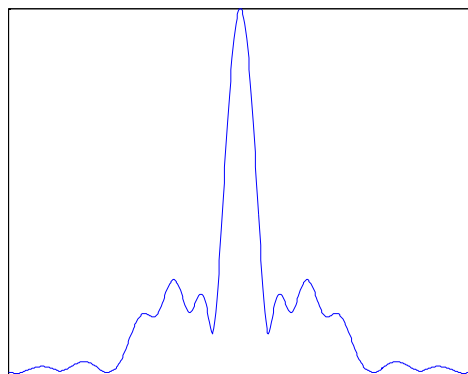
Simple solution?



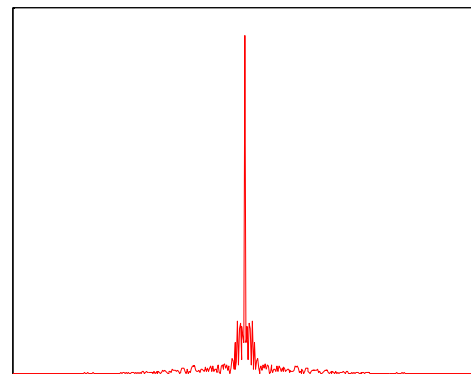
DFT domain (magnitude)



$\div$



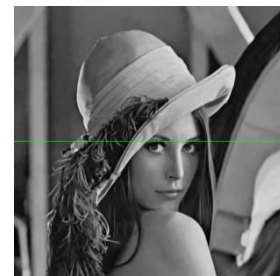
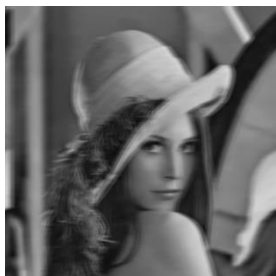
$=$



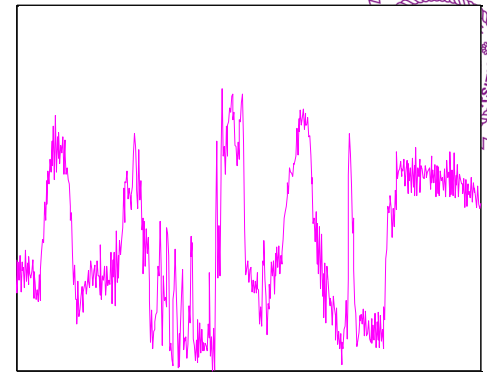
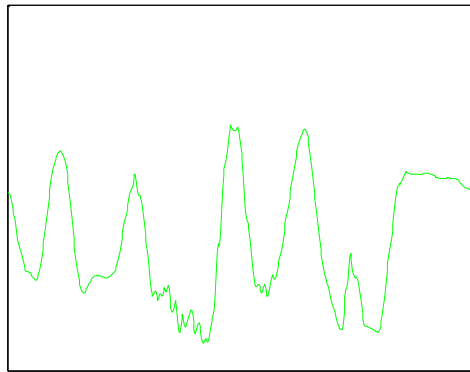
B

K

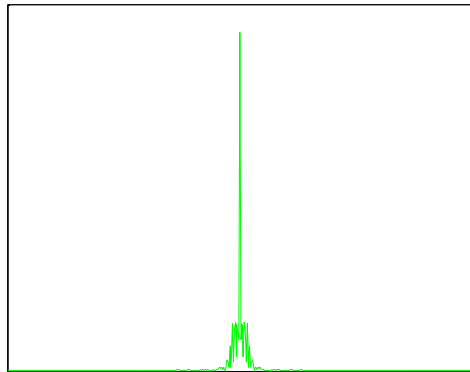
$\hat{I}$



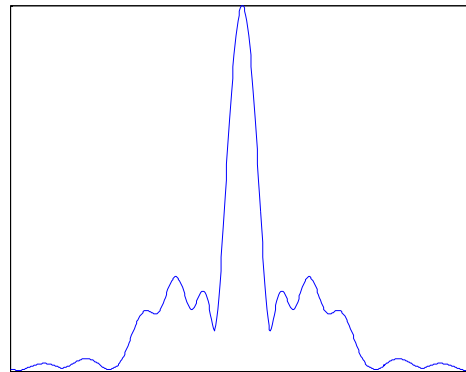
Instability



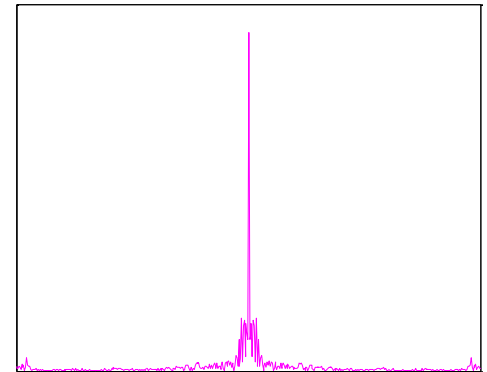
DFT domain (magnitude)



$\div$



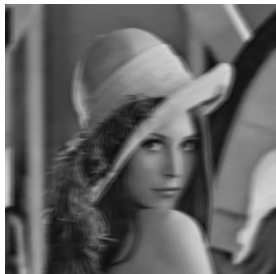
$=$



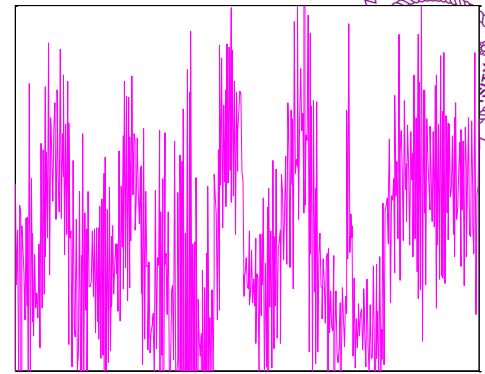
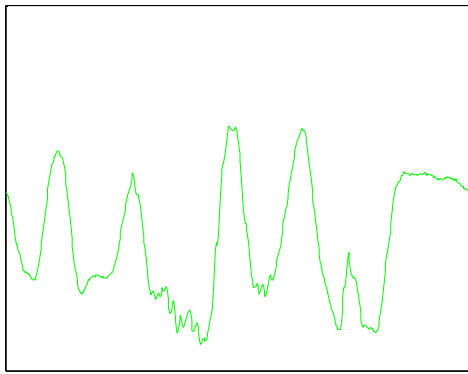
B (+AWGN $\sigma = 0.1$)

K

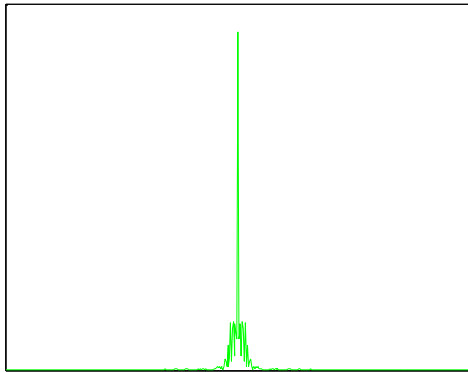
$\hat{I}$



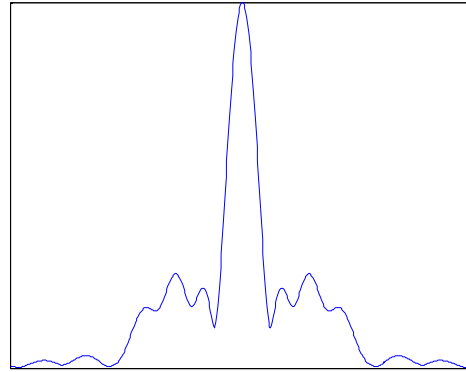
Instability



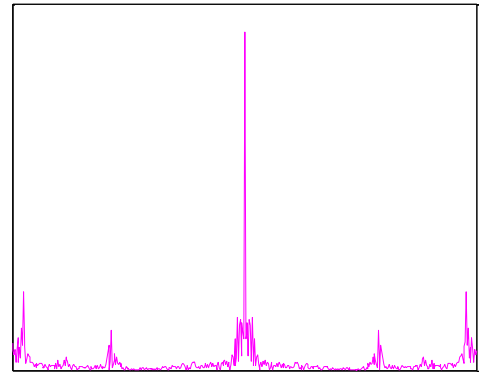
DFT domain (magnitude)



$\div$



$=$



B (+AWGN $\sigma = 0.5$)

K

$\hat{I}$

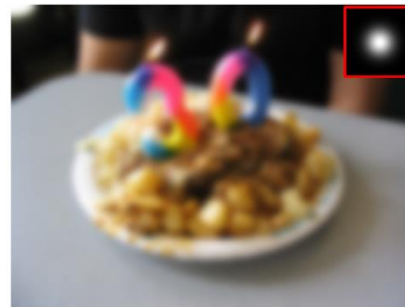


Similar inputs result in different outputs!



Difficulty of deblurring (1/3)

- For non-blind deconvolution (given kernel K)
 - K may have zeros in magnitude response
 - Some information of I is gone and can't be recovered
 - $FFT\{B\}/FFT\{K\}$ will amplify noise at frequencies with zeros
 - Deblurring quality is kernel-dependent
 - Good quality for sharp kernel (e.g. **camera shake**)
 - Blurred result for lowpass kernel (e.g. defocus blur)





Difficulty of deblurring (2/3)

- For blind deconvolution (find the kernel $\mathbf{K}$)
 - Multiple solutions of $\mathbf{K}$ are possible
 - Addressed by prior knowledge ($\mathbf{K}$ is sharp)
 - Real $\mathbf{K}$ may not be good for deconvolution (many zeros)
 - Convolution doesn't hold at saturation regions
 - No information for $\mathbf{K}$ in smooth texture regions



Difficulty of deblurring (3/3)

- Algorithm limit
 - To sum up, deblurring tries to restore the image as sharp as possible, not to recover it exactly
 - Shift-invariant camera shake is the major target here
 - Other blur kernels are more difficult: shift-variant kernel, motion blur, ...



Outline

- Single-image deblurring
 - **Non-blind deconvolution**
 - **Wiener deconvolution**
 - Richardson-Lucy (RL) deconvolution
 - Bilateral RL deconvolution
 - Multi-scale deconvolution
 - Total variation regularization
 - Blind deconvolution
- Multi-image deblurring

Wiener deconvolution

- Wiener filter for each frequency component

$$\hat{I} = B \cdot \frac{1}{K} \quad \Rightarrow \quad \hat{I}(f) = B(f) \cdot \frac{1}{K(f)} \left[\frac{1}{1 + \frac{1}{|K(f)|^2 \cdot \text{SNR}(f)}} \right]$$

instable

I

B

$\hat{I}$



credit: Wikipedia



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$Q: B, K \rightarrow I?$
 $\mathcal{H}: n$ is i.i.d. Poisson



Richardson-Lucy (RL) deconvolution

- Same iterative update method discovered independently by Richardson (1972) and Lucy (1974)
 - Both for astronomical images for which the kernel can be easily identified
 - But from different Bayesian models

$$I^{t+1} = I^t \left[K^* \otimes \frac{B}{(I^t \otimes K)} \right]$$

pixel-wise multiplication

pixel-wise division

$\int K(x) dx = 1$
normalized kernel
 $K^*(i, j) = K(-i, -j)$
adjoint kernel



Reasoning for RL deconvolution

- Assume shot noise with Poisson distribution
 - Especially true for astronomical images (few photons)
 - Assume the blurred signal $J(x)$ at position x is the expected value of a Poisson distribution:

$$p(B(x)|J(x)) = \frac{J(x)^{B(x)} \exp(-J(x))}{B(x)!}$$

Aggregate pixels for the blurred image

$$J(x) = (I \otimes K)(x)$$

$$p(B|I) = \prod_x \frac{(I \otimes K)(x)^{B(x)} \exp \{-(I \otimes K)(x)\}}{B(x)!}$$



Reasoning for RL deconvolution

- Formulate the energy E using negative log likelihood

$$E(I) = \sum \{(I \otimes K) - B \cdot \log[(I \otimes K)]\}$$

- Maximum Likelihood estimation (energy minimization)

$$\frac{\partial E}{\partial I(x)} = 0 \Rightarrow 1 - \left(K^* \otimes \frac{B}{I \otimes K} \right)(x) = 0 \Rightarrow \left(K^* \otimes \frac{B}{I \otimes K} \right)(x) = 1$$

- Formulated iterative update according to ML estimation

$$I^{t+1} = I^t \left[\underbrace{K^* \otimes \frac{B}{(I^t \otimes K)}}_{\text{per-pixel adaptive weight}} \right]$$

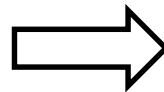
per-pixel adaptive weight

Properties of RL deconvolution

- Good properties for its performance
 - **Non-negativity**: I and B are non-negative
 - **Energy preserving**: the total energy of the image is preserved
- Convergence not proven but empirical
- Drawback: Ringing effect



RL deconv.



Close-up of ringing effect

- Observation: Propagated from edges to smooth regions
 - Occurs at smooth regions (small variance or gradient)
 - Occurs near strong edges (energy leak)
 - Ringing period is of similar size of the kernel





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$Q: B, K \rightarrow I?$
 $\mathcal{H}: n$ is i.i.d. Poisson
+ **edge-preserved smoothness**



Bilateral Richardson-Lucy (BRL) deconvolution

- Add edge-preserving regularization term $E_B(I)$
 - Make $I(x)$ and $I(y)$ closer only when their bilateral distance is small to preserve edges

$$I^* = \arg \min_I [E(I) + \lambda E_B(I)]$$

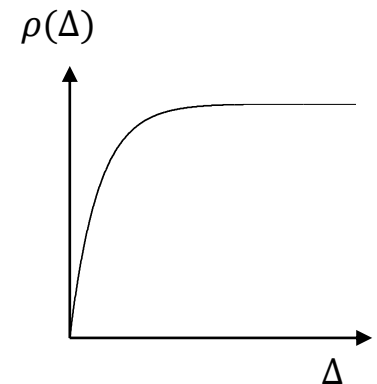
$$E_B(I) = \sum_x \sum_{y \in \Omega} f(|x - y|) \rho(|I(x) - I(y)|)$$

Spatial penalty

$$f(|x - y|) = \exp \left(-\frac{|x - y|^2}{2\sigma_s} \right)$$

Robust range penalty

$$\rho(|I(x) - I(y)|) = 1 - \exp \left(-\frac{|I(x) - I(y)|^2}{2\sigma_r} \right)$$



BRL derivation

(modified for clarity)

- Maximum Likelihood estimation (energy minimization)

$$\frac{\partial E_{BRL}}{\partial I(x)} = 0 \Rightarrow 1 - \left(K^* \otimes \frac{B}{I \otimes K} \right) (x) + \lambda \frac{\partial E_B}{\partial I(x)} = 0$$

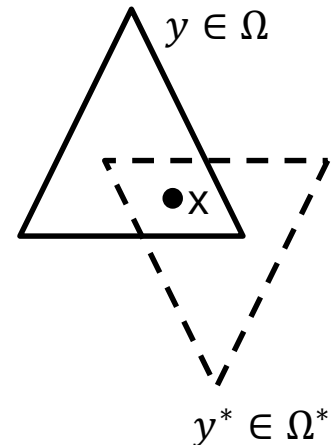
$$\Rightarrow I^{t+1} = \frac{I^t}{1 + \lambda \cdot \nabla E_B(I^t)} \left(K^* \otimes \frac{B}{I \otimes K} \right) (x)$$

- Derive the intensity gradient of $E_B(I)$

$$\frac{\partial}{\partial I(x)} f(x-y) \rho(I(x) - I(y)) \Rightarrow \begin{cases} I_y^d(x) = f(|x-y|) g(|I(x) - I(y)|) \cdot \frac{(I(x) - I(y))}{\sigma_r} \\ g(|I(x) - I(y)|) = \exp\left(-\frac{|I(x) - I(y)|^2}{2\sigma_r}\right) \end{cases}$$

$$\frac{\partial E_B}{\partial I(x)} = \sum_{y \in \Omega} I_y^d(x) + \sum_{y^* \in \Omega^*} I_{y^*}^d(x)$$

If Ω is isotropic (e.g. circle), $\Omega = \Omega^*$.



BRL result

- For 11x11 kernel

RL



- For 40x40 kernel
 - Ringing still exists





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 - **Multi-scale deconvolution**
 - Total variation regularization
 - Blind deconvolution
- Multi-image deblurring

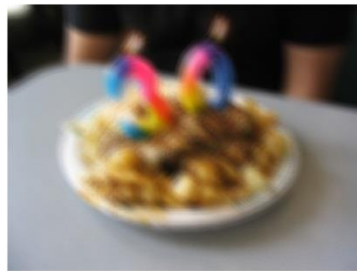
Q: $B, K \rightarrow I$?

$\mathcal{H}$: n is Poisson

+ edge-preserved
smoothness

+ **progressive
update** (ringing suppression)

Progressive deconvolution



blurred image



kernel



standard RL



true image

Progressive inter-scale deconvolution



scale 0



scale 2



scale 4



scale 6

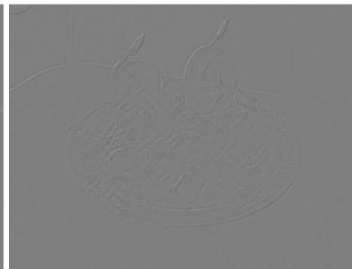
Progressive intra-scale deconvolution



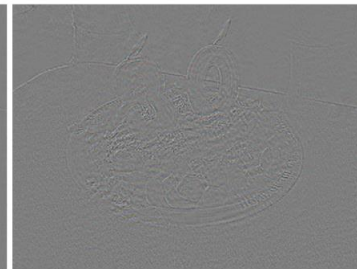
guide image



detail layer (1)



detail layer (2)



detail layer (3)

Progressive inter-scale deconvolution

- Joint BRL (JBRL) deconvolution
 - Use small-scale image as the guided image I^g to further regularize the energy

$$E_{JB}(I; I^g) = \sum_x \sum_{y \in \Omega} f(|x - y|) g'(|I^g(x) - I^g(y)|) \rho(|I(x) - I(y)|)$$

$$g'(|I^g(x) - I^g(y)|) = \exp \left(-\frac{|I^g(x) - I^g(y)|^2}{2\sigma_r^g} \right)$$

$$I_y^d(x) = f(|x - y|) g(|I(x) - I(y)|) \cdot g'(|I^g(x) - I^g(y)|) \cdot \frac{(I(x) - I(y))}{\sigma_r}$$

BRL

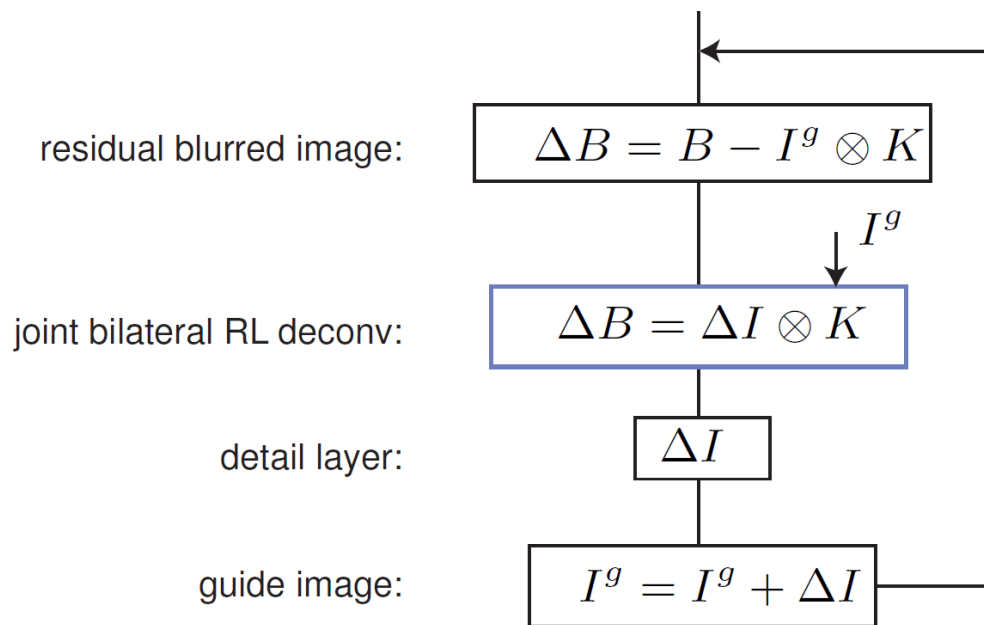


JBRL



Progressive intra-scale deconvolution

- Iterative residue deconvolution
 - Deconvolute residue, instead of high-intensity image, to reduce the ringing effect
 - Iteratively update guide/residue images

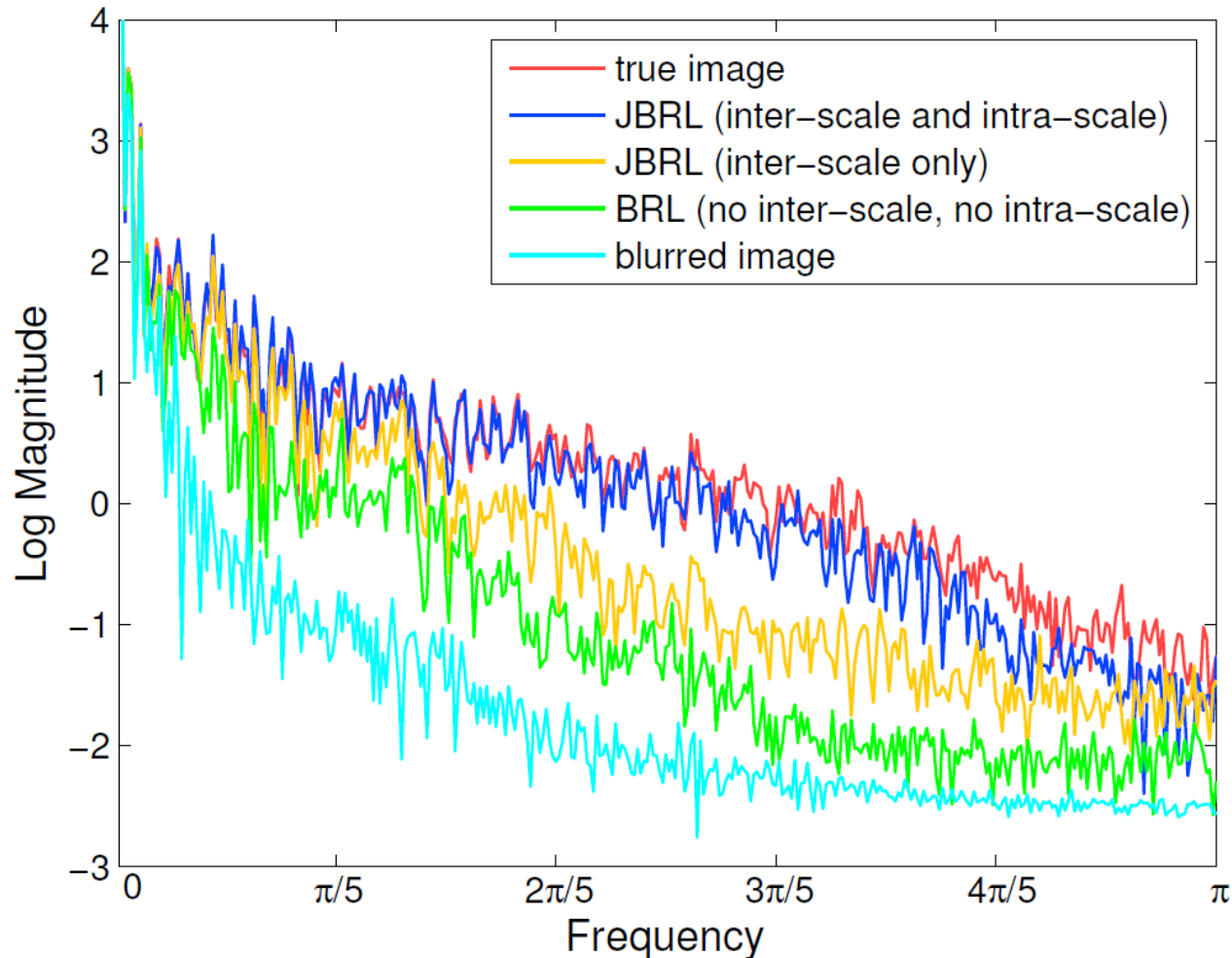


Inter/intra



Experimental results

- DFT curves of a 1-D scanline



Experimental results

RL

Progressive





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 - Non-blind deconvolution
 - Wiener deconvolution
 - Richardson-Lucy (RL) deconvolution
 - Bilateral RL deconvolution
 - Multi-scale deconvolution (ringing suppression)
 - **Total variation regularization**
 - Blind deconvolution
- Multi-image deblurring

$Q: B, K \rightarrow I?$

$\mathcal{H}: \nabla I$ should be small



Total variation regularization

- TVL1 deconvolution
 - L1-norm data term (first term) is more robust than L2 norm
 - Total-variation regularization term (second term) constrains the (isotropic) variation of gradients
 - Solved by iterative algorithms, e.g. half-quadrature splitting

$$E(I) = \|I \otimes k - B\| + \lambda \|\nabla I\|_{tv}$$

$$\|\nabla I\|_{tv} = \sum_p \sqrt{I(p)_x^2 + I(p)_y^2}$$



Outline

- Single-image deblurring

- Non-blind deconvolution

$Q: B, K \rightarrow I?$

$\mathcal{H}: \nabla I$ should be small

- **Blind deconvolution**

- **Gradient-based kernel estimation**

- Variational Bayesian method [appendix]

- Multi-image deblurring

$Q: B \rightarrow K?$

$\mathcal{H}: n$ is AWGN

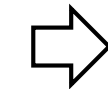
+ MoG gradient prior

+ MoE kernel prior

S: Variational Bayesian,
not MAP

Gradient-based kernel estimation

- Formulate the problem using image gradient and solve it using variational Bayesian method
 - Image gradients behave more predictably
 - Variational Bayesian *approximates* the full posterior distribution and estimates by maximizing marginal probability
 - Direct MAP minimizes gradients to zeros to have roughly flat images
- Multi-scale approach for large kernel size
- Use RL deconvolution
 - Will be better to use progressive deconvolution (2008)

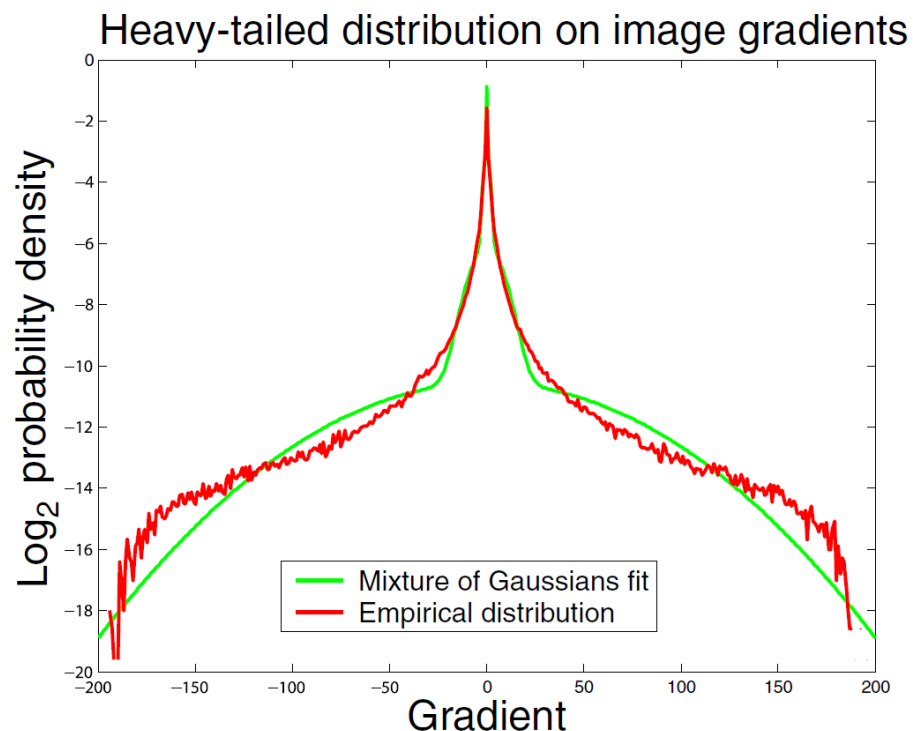


K

Ref : R. Fergus, et. al., "Removing camera shake from a single photograph," SIGGRAPH'06.

Image gradient prior

- Heavy-tailed distribution fit by mixture-of-Gaussians

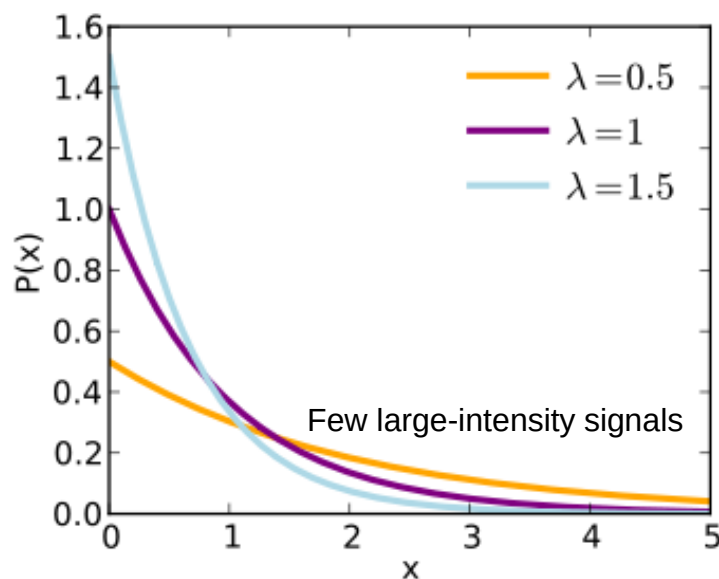


$$p(\nabla \mathbf{L}_p(i)) = \sum_{c=1}^C \pi_c \mathcal{N}(\nabla \mathbf{L}_p(i) | 0, v_c)$$

$\nabla \mathbf{L}_p(i)$: image gradient at position i in the image patch $\mathbf{L}_p$
 c : mixture component index

Kernel prior

- Sparse and positive distribution fit by mixture-of-Exponential



$$p(\mathbf{K}_j) = \sum_{d=1}^D \pi_d \mathbf{E}(\mathbf{K}_j | \lambda_d)$$

$\mathbf{K}_j$: blur kernel element at position j
 d : mixture component index

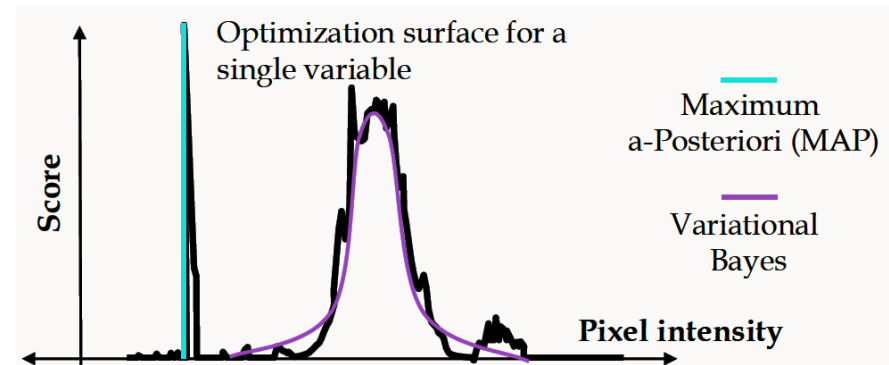
Bayesian formulation

$$\begin{aligned}
 \underset{\text{posterior}}{p(\mathbf{K}, \nabla \mathbf{L}_p | \nabla \mathbf{P})} &\propto \overset{\text{Likelihood}}{p(\nabla \mathbf{P} | \mathbf{K}, \nabla \mathbf{L}_p)} \overset{\text{Image prior}}{p(\nabla \mathbf{L}_p)} \overset{\text{Kernel prior}}{p(\mathbf{K})} \\
 &= \prod_i \mathcal{N}(\nabla \mathbf{P}(i) | (\mathbf{K} \otimes \nabla \mathbf{L}_p(i)), \sigma^2) \\
 &\quad \prod_i \sum_{c=1}^C \pi_c \mathcal{N}(\nabla \mathbf{L}_p(i) | 0, v_c) \prod_j \sum_{d=1}^D \pi_d \mathcal{E}(\mathbf{K}_j | \lambda_d)
 \end{aligned}$$

- MAP by conjugate gradient descent $\Rightarrow$ **Failed!**

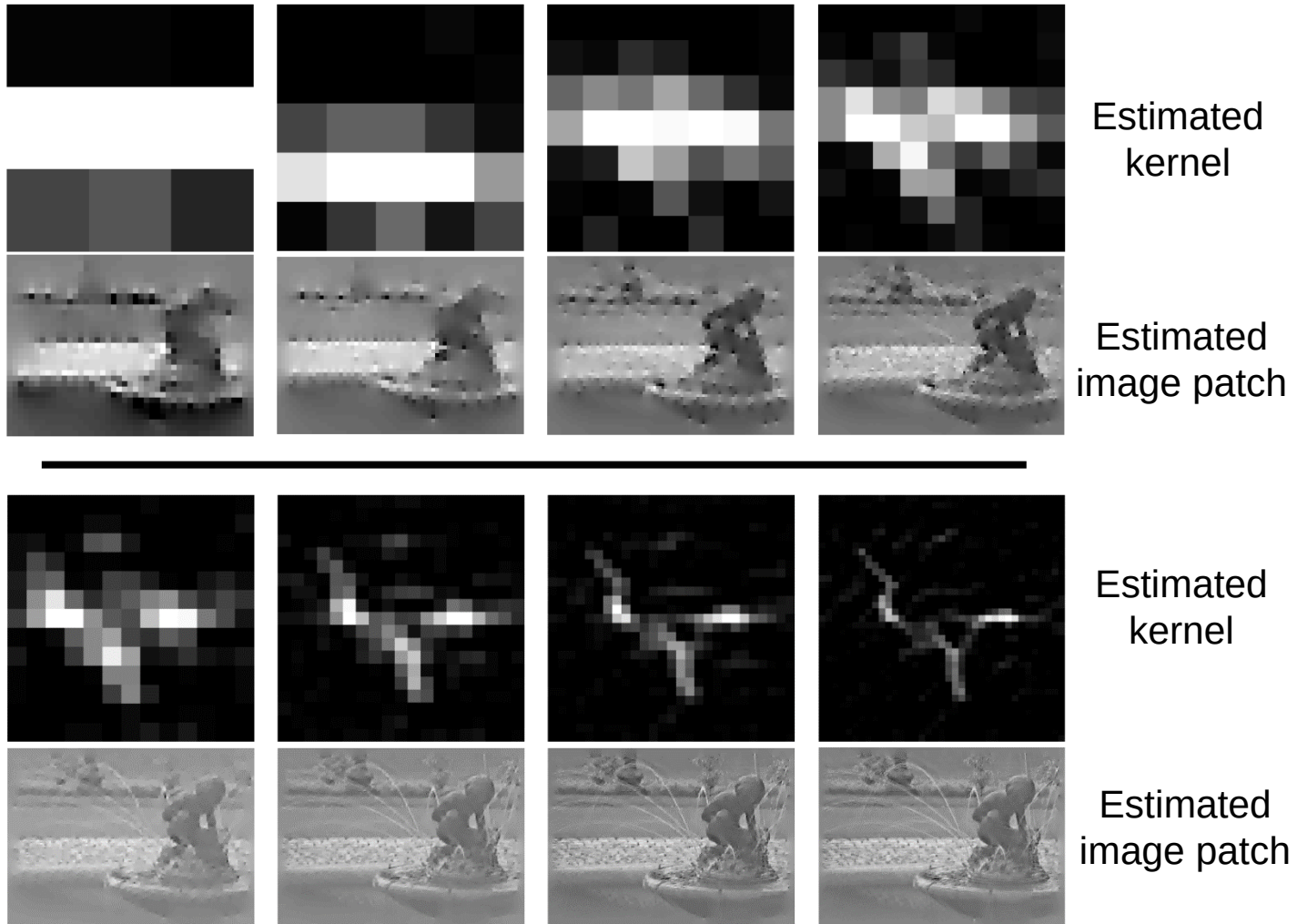
$\Rightarrow$ Variational Bayesian!

- Fit pixel-wise and kernel-wise distributions;
- use expectation to estimate kernel;
- finally perform non-blind deconvolution

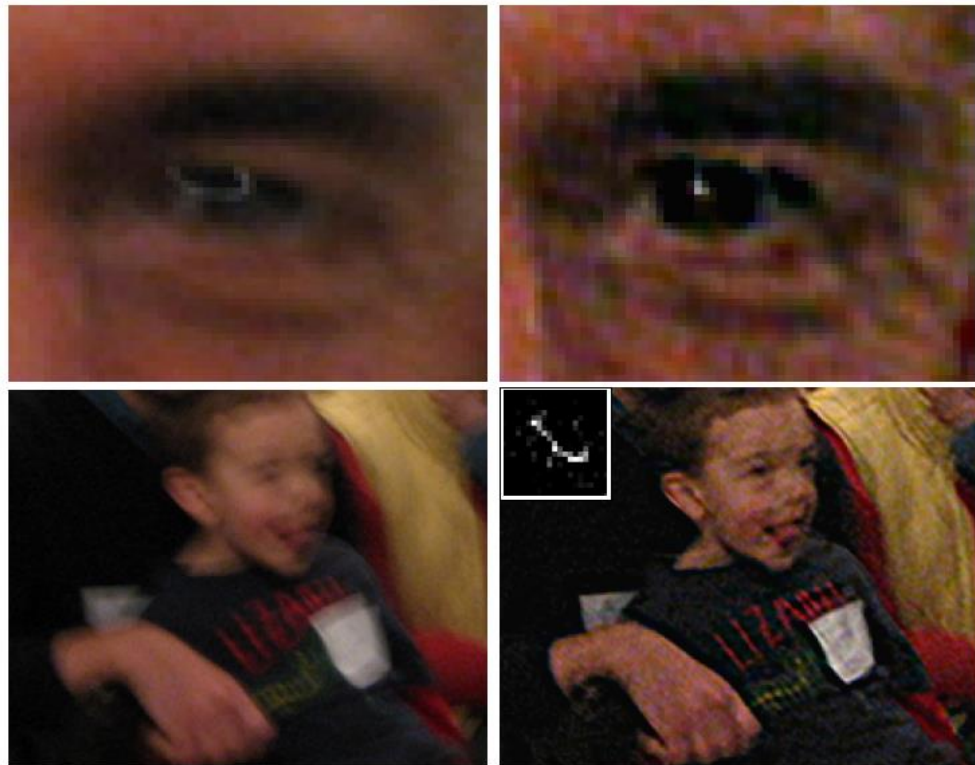


Multi-scale approach

- Use previous-scale result as initialization



Blind deconvolution result



RL deconvolution can be
replaced for better results





Outline

- Single-image deblurring
 - Non-blind deconvolution
 - Blind deconvolution

Q: $B, K \rightarrow I$?

Q: $B \rightarrow K$?; $\mathcal{H}$: priors

- **Multi-image deblurring**
 - **Image-pair deblurring**
 - Burst deblurring

Q: $B, I_{noisy} \rightarrow I$?
 $\mathcal{H}$: $\text{Denoise}(I_{noisy}) \cong I$

Image Deblurring with Blurred/Noisy Image Pairs

- Use the denoised noisy image as the base and perform iterative residue deconvolution
- Use a simple least-squares optimization to estimate the kernel



(a) blurred image

Shutter 1s, ISO 100



(b) noisy image

Shutter 0.01s, ISO 1600



(c) enhanced noisy image

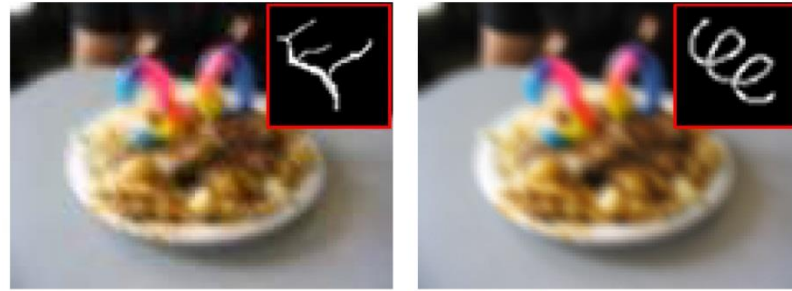


(d) our deblurred result

Ref : L. Yuan, et. al., "Image deblurring with blurred/noisy image pairs," SIGGRAPH'07.

Comparison

- Faster and better blind deconvolution can be achieved using image pairs



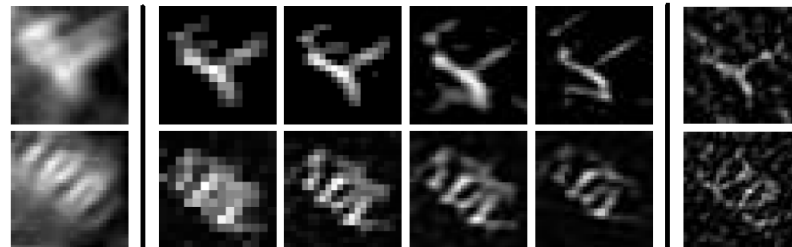
(a) blurry images and true kernels



(b) noisy image



(c) denoised image



(d)

(e) Results by Fergus et.al.

(f)



(g)

(h) Our Results

(i)



Outline

- Single-image deblurring
- Multi-image deblurring
 - Image-pair deblurring
 - **Burst deblurring**

Q: $B, I_{noisy} \rightarrow I$?

Q: $B_1, B_2, \dots, B_N \rightarrow I$?
 $\mathcal{H}: K_n$ is random



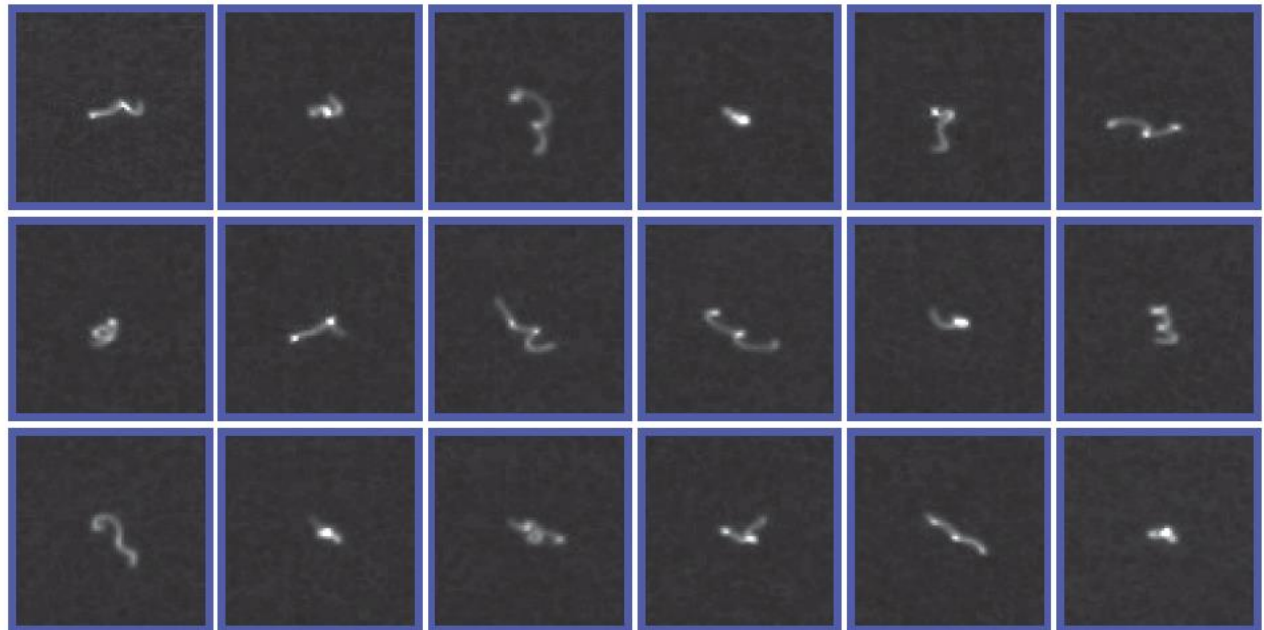
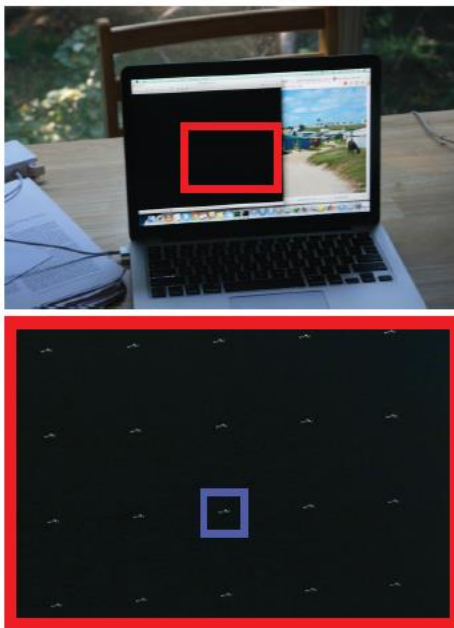
Burst deblurring

- Accumulation of burst images in Fourier domain
 - **Benefit:** No inverse problem solving
 - e.g. ~~kernel estimation, deconvolution~~
 - **Requirement:** Burst image capturing
 - **Assumption:** Kernels in each image are *random*, and each kernel *disperses* the spectrum and preserves energy

Utilize the random behavior of burst kernels as an implicit clue to aggregate images.

Random nature of hand tremor

- Low correlation between successive frames for low shutter speed
 - Tremor source: arm ($<5\text{Hz}$), wrist ($5\text{-}20\text{Hz}$) and fingers ($20\text{-}30\text{Hz}$)

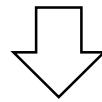


Hand tremor in successive shoots



Blurring kernel doesn't amplify spectrum

$$k(\mathbf{x}) \geq 0 \text{ and } \int k(\mathbf{x}) = 1$$



$$|\hat{k}(\zeta)| = \left| \int k(\mathbf{x}) e^{i\mathbf{x} \cdot \zeta} d\mathbf{x} \right| \leq \int |k(\mathbf{x})| d\mathbf{x} = \int k(\mathbf{x}) d\mathbf{x} = 1$$

- The attenuation of each spectrum coefficient is random
 - Aggregate large amplitudes across blurred images for accurate estimation
- $\Rightarrow$ Fourier burst accumulation!**



Fourier burst accumulation (FBA)

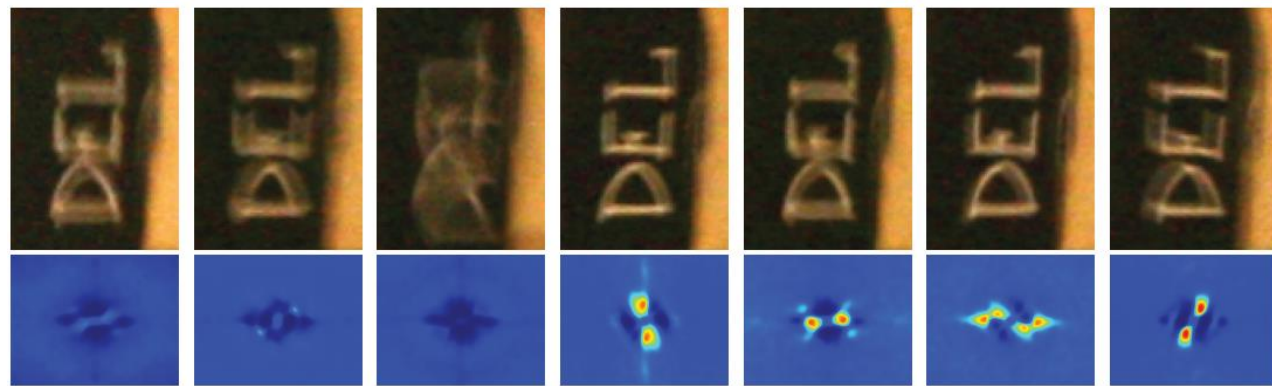
- Perform robust weighted average for each frequency component ζ in Fourier domain

$$u_p(\mathbf{x}) = \mathcal{F}^{-1} \left(\sum_{i=1}^M w_i(\zeta) \cdot \hat{v}_i(\zeta) \right) (\mathbf{x})$$

$$w_i(\zeta) = \frac{|\hat{v}_i(\zeta)|^p}{\sum_{j=1}^M |\hat{v}_j(\zeta)|^p},$$

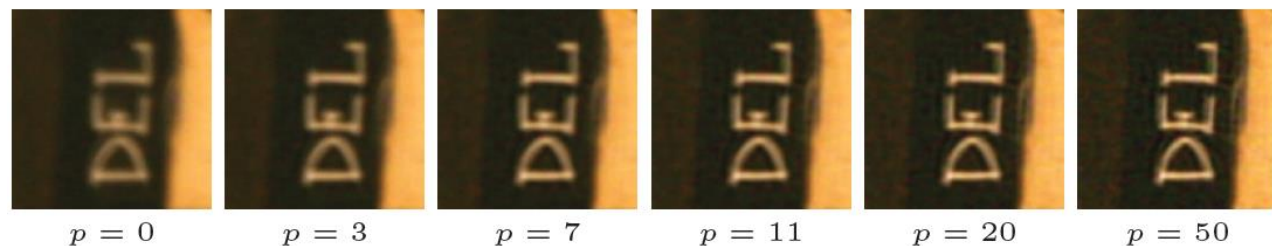

Control the accumulation behavior:
 $p = 0$: simple average;
 $p \rightarrow \infty$: maximum pooling.

Behavior of parameter p

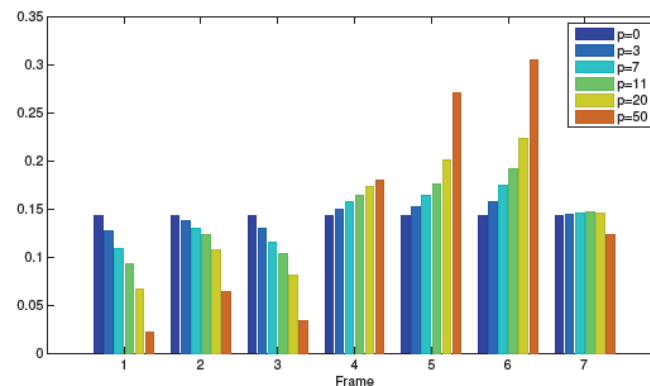


(a) Frames crop 1-7 and the Fourier weights w_i for $p = 11$.

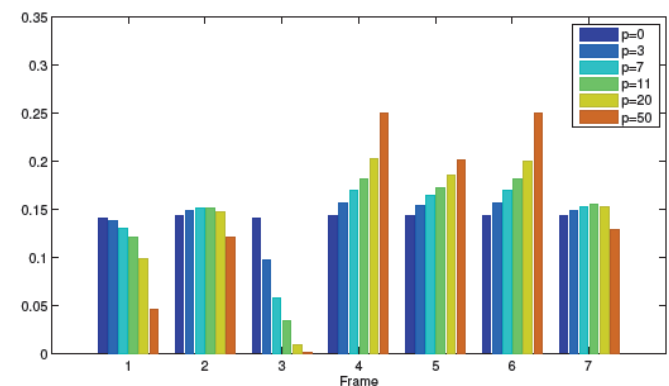
$p = 11$ selected



(b) Fourier Aggregation results for different p values.



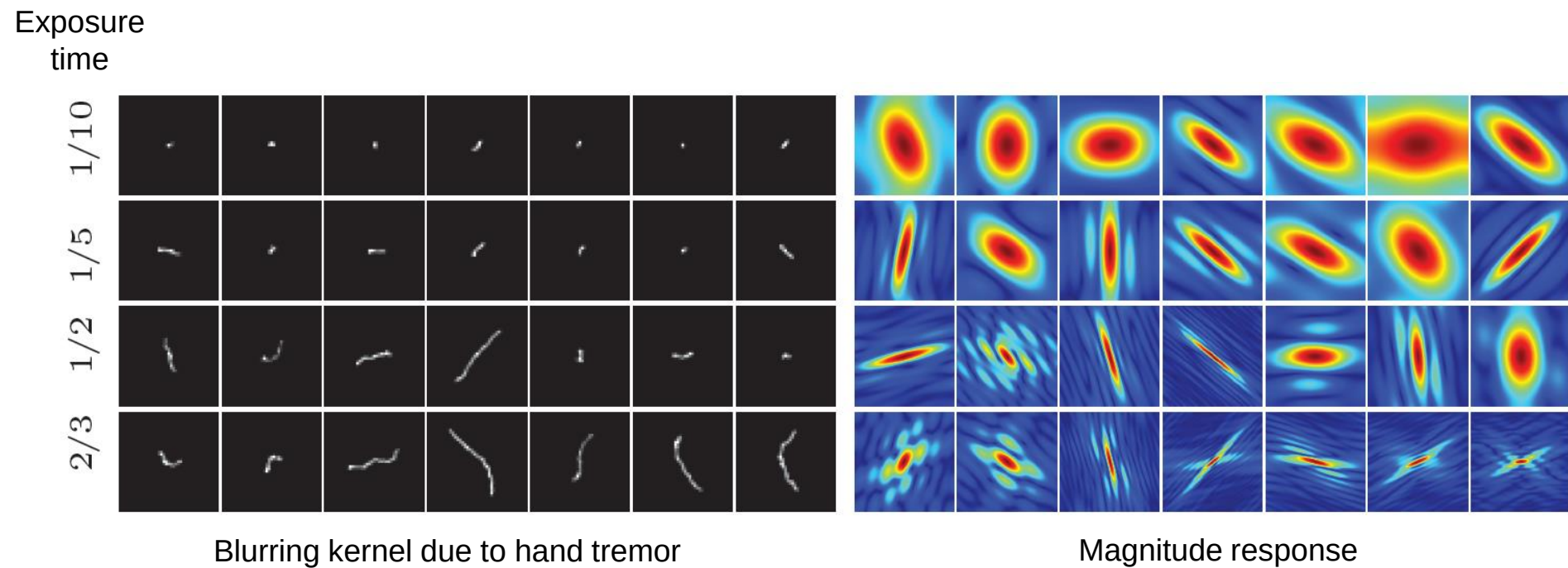
(c) Weights energy distribution
 $w_i(\zeta)$



(d) Weighted frames energy distribution
 $w_i(\zeta) \cdot \hat{v}_i(\zeta)$

Behavior of $\hat{k}(\zeta)$

- If hand tremor is random enough, Fourier burst accumulation can cover most of frequency components (red in the below)



Experimental results

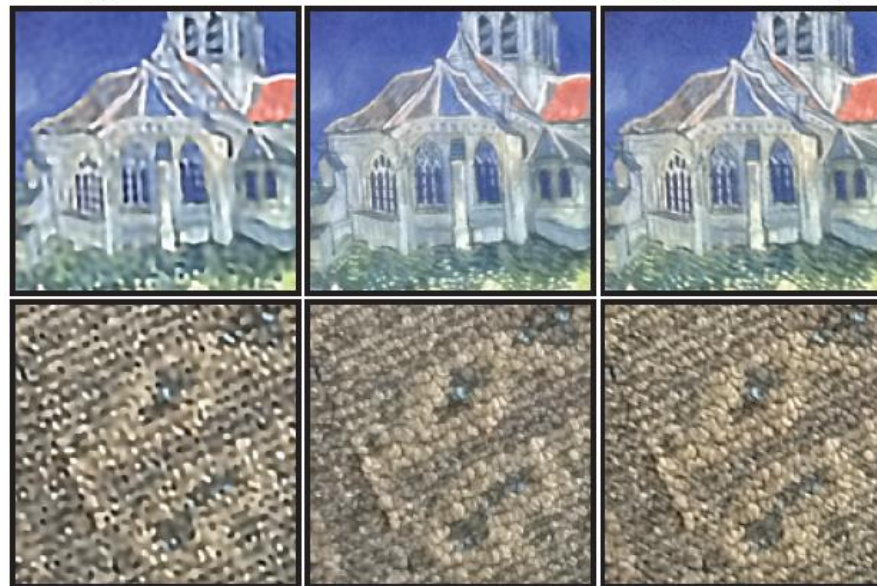
- 8~13 burst images
- 6~8M pixels



Typical Shot

Best Shot

Align and average



Šroubek &
Milanfar [25]

Zhang *et al.* [30]

proposed method

several hours few seconds for FBA



Typical Shot

Best Shot

Align and average

Šroubek &
Milanfar [25]

Zhang *et al.* [30]

proposed method
(no final sharp.)

proposed method

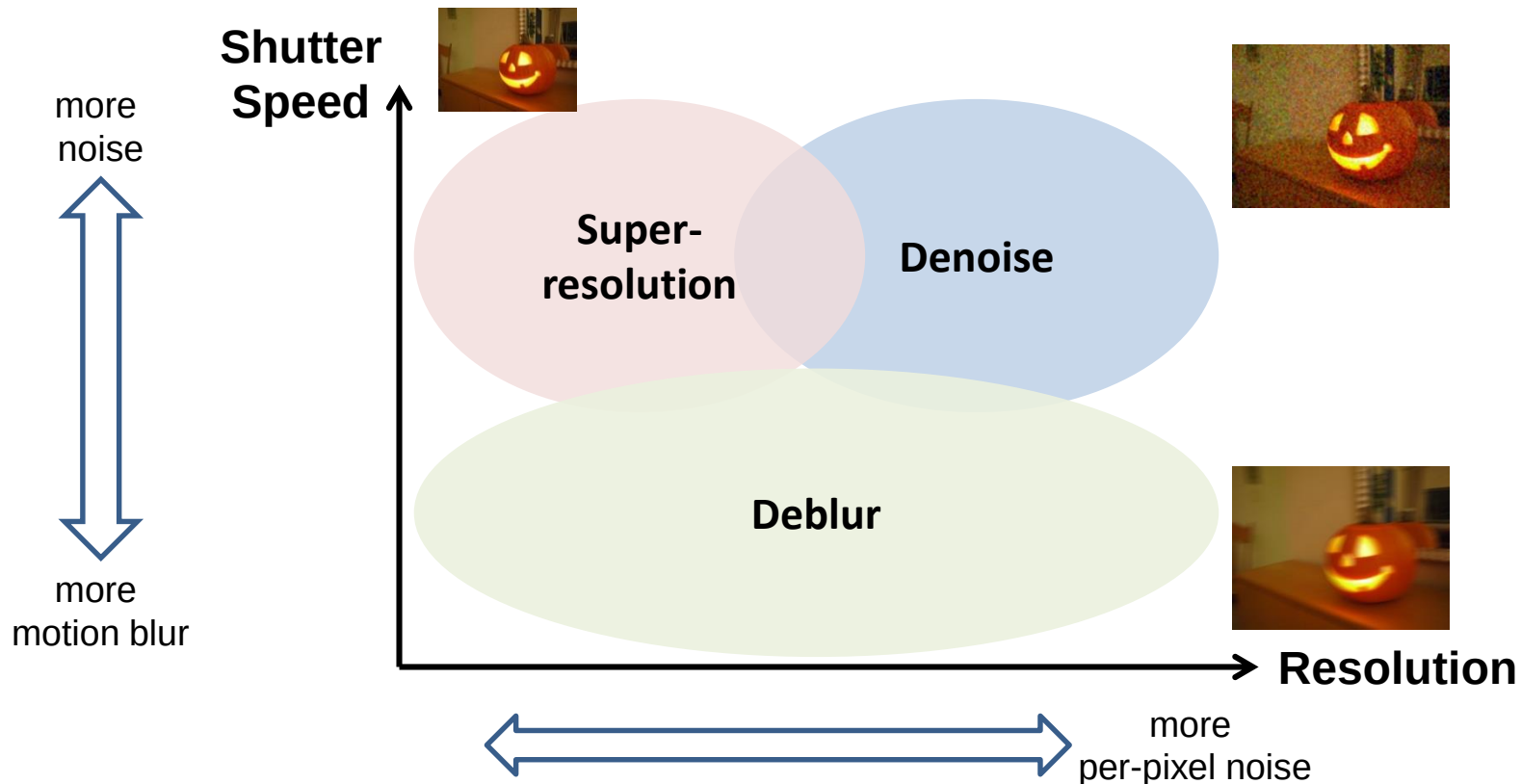


Implementation of burst deblurring

- Hardware-oriented solution for deblurring
 - Use physical facilities (i.e. burst) to ease the problem
 - Computation load is small
 - Burst shots widely available
 - FFT is hardware-friendly
- Image alignment is an issue
 - For averaging Fourier coefficients, the images have to be aligned well.
 - This work uses SIFT for alignment (time-consuming!)
 - But gyro sensor doesn't help for its slow and delayed response

Notes on image deblurring

- Should work in linear domain based on its assumption
- Collaborative or competitive for image restoration?

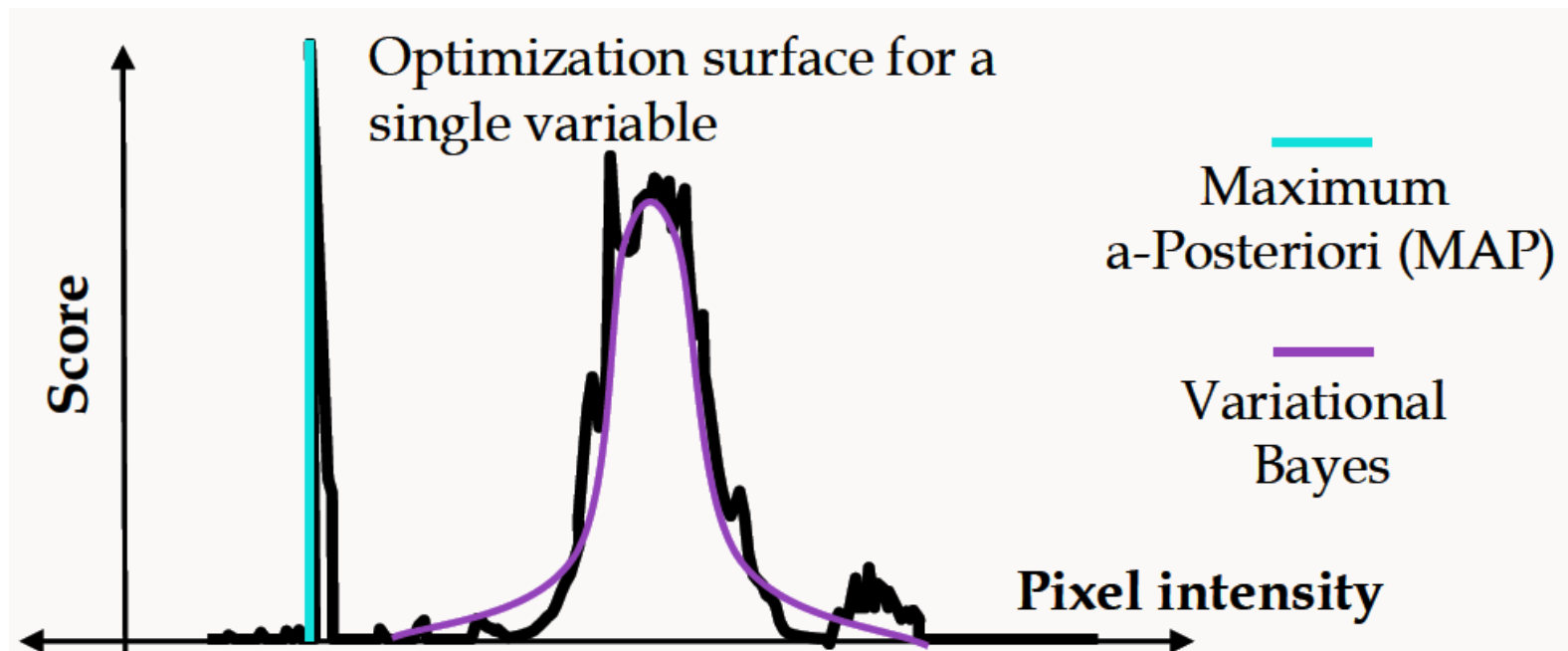




Appendix

Variational Bayesian approach

- Approximate the *intractable* joint distribution using *tractable* independent distributions
- Keep track of the tractable distribution of each variable, instead of only a single estimate





Variational Bayesian (Ensemble learning)

- Approximate a complex $P(\mathbf{z})$ by a simpler $Q(\mathbf{z})$ to minimize the KL divergence (KLD)

- $\mathbf{z} = \{z_1, z_2, \dots, z_n\}$: unobserved random variables

$$D_{KL}(Q \parallel P) = E_Q \left[\log \frac{Q}{P} \right] = \left\langle \log \frac{Q}{P} \right\rangle_Q = \int \underbrace{\log \frac{Q}{P}}_{\text{Lagrangian}} \cdot \underbrace{Q dz}_{\text{Functional of } Q} = J[Q]$$

- Let $Q(\mathbf{z}) = \prod_i q_i(z_i)$ and perform variational analysis for q_i
 - Assume $q_{j \neq i}$ are fixed

$$\frac{\partial L}{\partial q_i} - \frac{d}{dz_i} \frac{\partial L}{\partial q'_i} = 0 \Rightarrow \log q_i = \mathbf{E}_{q_{j \neq i}} [\log P(\mathbf{z})] + \underbrace{\text{const.}}_{\text{(for pdf normalization)}}$$

Calculus of variations
(Euler-Lagrange equation)

- Ensemble learning

- Update all $q_i^{(t+1)}$ from $q_i^{(t)}$ iteratively

$$\log q_i^{(t+1)}(z_i) = \mathbf{E}_{q_{j \neq i}^{(t)}} [\log P(\mathbf{z})] + \text{const.}$$

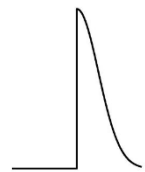


Useful results. :)

On gradient-based kernel estimation

Posterior $P(\mathbf{K}, \nabla \mathbf{L}_p | \nabla \mathbf{P})$

approximated by $\prod_{j,i} q(\mathbf{K}_j) \cdot q(\nabla \mathbf{L}_p(i)) \cdot q(\sigma^{-2})$



Non-negative
rectified Gaussian

Gaussian

Gamma distribution
(fit to image noise)

Cost function (KLD)

$$\left\langle \log \frac{q(\nabla \mathbf{L}_p)}{p(\nabla \mathbf{L}_p)} \right\rangle_{q(\nabla \mathbf{L}_p)} + \left\langle \log \frac{q(\mathbf{K})}{p(\mathbf{K})} \right\rangle_{q(\mathbf{K})} + \left\langle \log \frac{q(\sigma^{-2})}{p(\sigma^2)} \right\rangle_{q(\sigma^{-2})}$$

- Through some simplification, each ensemble learning update can be replaced by the update of the derived q -functions' parameters
- The kernel is estimated by the mean $\langle \mathbf{K} \rangle_{q(\mathbf{K})}$